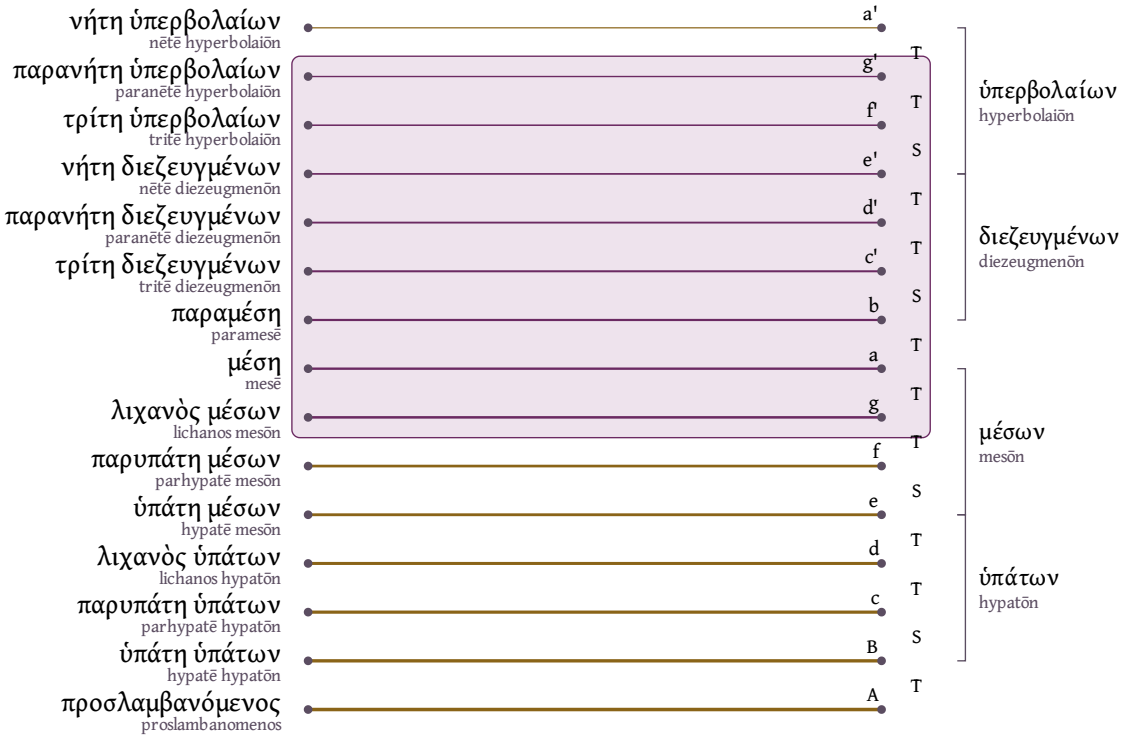
ὑποφρύγιον Hypophrygian

Ratio model: 9:8 and 256:243. Mesē is held at 330 Hz. Selected string: τρίτη ὑπερβολαίων, 521.4815 Hz.



T = tone, 203.910 cents. S = small step, 90.225 cents. Two S steps are 23.460 cents smaller than T; “semitone” does not mean half of T in this ratio model.

The highlighted span marks the selected octave. Equal row spacing orders notes; it does not measure pitch distance or physical string lengths.

Hear the octave, then return

Eight ascending notes followed by seven descending notes. Synthetic harmonic plucks, 44,100 Hz mono PCM. Each note lasts 1.6 seconds; onsets are 0.42 seconds apart. Tails overlap as in the workshop.

Fifteen named positions; the band marks Cleonides’ Hypophrygian.

String, from low pitch	Hz	Cents from mesē
προσλαμβανόμενος proslambanomenos	165.0000	-1200.000
ὑπάτη ὑπάτων hypatē hypatōn	185.6250	-996.090
παρυπάτη ὑπάτων parhypatē hypatōn	195.5556	-905.865

String, from low pitch	Hz	Cents from mesē
λιχανὸς ὑπάτων lichanos hypatōn	220.0000	-701.955
ὑπάτη μέσων hypatē mesōn	247.5000	-498.045
παρυπάτη μέσων parhypatē mesōn	260.7407	-407.820
λιχανὸς μέσων lichanos mesōn	293.3333	-203.910
μέση mesē	330.0000	0.000
παραμέση paramesē	371.2500	203.910
τρίτη διεζευγμένων tritē diezeugmenōn	391.1111	294.135
παρανήτη διεζευγμένων paranētē diezeugmenōn	440.0000	498.045
νήτη διεζευγμένων nētē diezeugmenōn	495.0000	701.955
τρίτη ὑπερβολαίων tritē hyperbolaiōn	521.4815	792.180
παρανήτη ὑπερβολαίων paranētē hyperbolaiōn	586.6667	996.090
νήτη ὑπερβολαίων nētē hyperbolaiōn	660.0000	1200.000

Read the named span

> ἕκτον … ἔστι δὲ ἀπὸ λιχανοῦ μέσων ἐπὶ παρανήτην ὑπερβολαίων, ἐκαλεῖτο δὲ ὑποφρύγιον.
>
> The sixth … runs from lichanos mesōn to paranētē hyperbolaiōn; it was called Hypophrygian.
>
> *Cleonides, Harmonic Introduction 9; Jan 1895, p. 198*

The half-step positions and named endpoints follow Cleonides. The numerical pitches, reference mesē and synthetic timbre are separate modern teaching choices. The equal-step option realizes tones as 200 cents and half-tones as 100; it does not establish that Aristoxenus prescribed modern equal temperament. The ratio option uses 9:8 and 256:243. White-key letters identify positions, not equal-tempered frequencies. No ancient instrument, absolute pitch, melody or performance is reconstructed.

Cleonides in Jan (1895), complete printed p. 197. Public domain. Source image.

ἐστι σχῆμα οὗ πρῶτον τὸ ἡμιτόνιον ἐπὶ τὸ βαρὺ κεῖται, δεύτερον δὲ οὗ πρῶτον ἐπὶ τὸ ὀξύ, τρίτον οὗ δεύτερον ἐπὶ τὸ ὀξύ, τέταρτον οὗ πρῶτον ἐπὶ τὸ βαρύ.

Τοῦ δὲ διὰ πασῶν εἴδη ἐστὶν ἑπτά. πρῶτον μὲν τὸ ὑπὸ βαρυπύκνων περιεχόμενον, οὗ πρῶτος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ ὑπάτης ὑπάτων ἐπὶ παραμέσην, ἐκαλεῖτο δὲ ὑπὸ τῶν ἀρχαίων μιξολύδιον.

Δεύτερον τὸ ὑπὸ μεσοπύκνων περιεχόμενον οὗ δεύτερος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ παρυπάτης ὑπάτων ἐπὶ τρίτην διεζευγμένων, ἐκαλεῖτο δὲ λύδιον.

Τρίτον τὸ ὑπὸ ὀξυπύκνων περιεχόμενον, οὗ τρίτος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ λιχανοῦ ὑπάτων ἐπὶ παρανήτην διεζευγμένων, ἐκαλεῖτο δὲ φρύγιον.

Τέταρτον τὸ ὑπὸ βαρυπύκνων περιεχόμενον, οὗ

---

1 *οὗ πρῶτον* — 3 *δεύτ. ἐπὶ τὸ ὀξύ* om. M¹N. *ἐπὶ τὸ βαρύ* om. L. *κεῖται* W**B**, om. rel. 2 *τρίτον οὗ* — *ὀξύ* om. WL. 3 *τέταρτον οὐ* M. *πρῶτον ἐπὶ*] *τρίτον ἐπὶ* libri, cf. 196, 18. *βαρύ* libb., *ὀξύ* Meib. 6 *ἔστι δὲ*] ἢ W. 8 *δεύτερον δὲ τὸ* L. 12 *ἀπὸ*] *ὑπὸ* L. 13 ad 198, 14 scholion *τὸ δὲ ὑπὸ διατόνου ἤτοι λιχανοῦ φρύγιον.* M².

Dia pason.

Forma 1.	Hc	d	ef	g	a — h				mixolydia.
„ 2.	c	d	ef	g	a — hc′				lydia
„ 3.		d	ef	g	a — hc′	d′			phrygia
„ 4.	doria		**ef**	**g**	**a — hc′**	**d′**	**e′**		doria
„ 5.	hypolydia		f	g	a — hc′	d′	e′f′		
„ 6.	hypophrygia			g	a — hc′	d′	e′f′	g′	
„ 7.	hypodoria				a — hc′	d′	e′f′	g′	a′.

De his modis agit Bellermann Anon. p. 9. 35. Tonleitern p. 7. Gevaert Histoire I 127. Jan ad Bacchium p. 10. Westphal Harmonik³ p. 136. quae tamen hic de dynami sonorum somniavit, ea refelli in Berliner Wochenschrift 1886, p. 1134. et quae de variorum modorum exitu statuerat Gevaertusque ab illo recepit, ea falsa esse W. declaravit in Aristoxeno (1893) p. 1.

Cleonides in Jan (1895), complete printed p. 198. Public domain. Source image.

τέταρτος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ ὑπάτης μέσων ἐπὶ νήτην διεζευγμένων, ἐκαλεῖτο δὲ δώριον.

Πέμπτον τὸ ὑπὸ μεσοπύκνων περιεχόμενον οὗ πέμπτος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ παρυπάτης (p. 16) μέσων ἐπὶ τρίτην ὑπερβολαίων, ἐκαλεῖτο δὲ ὑπολύδιον.

Ἕκτον τὸ ὑπὸ ὀξυπύκνων περιεχόμενον, οὗ ἕκτος ὁ τόνος ἐπὶ τὸ ὀξύ· ἔστι δὲ ἀπὸ λιχανοῦ μέσων ἐπὶ παρανήτην ὑπερβολαίων, ἐκαλεῖτο δὲ ὑποφρύγιον.

Ἕβδομον τὸ ὑπὸ βαρυπύκνων περιεχόμενον, οὗ πρῶτος ὁ τόνος ἐπὶ τὸ βαρύ· ἔστι δὲ ἀπὸ μέσης ἐπὶ νήτην ὑπερβολαίων ἢ ἀπὸ προσλαμβανομένου ἐπὶ μέσην· ἐκαλεῖτο δὲ κοινὸν καὶ λοκριστὶ καὶ ὑποδώριον.

Ἐν δὲ διατόνῳ πρῶτον μέν ἐστιν εἶδος τὸ διὰ πασῶν, οὗ πρῶτον μὲν ἐπὶ τὸ βαρύ, τέταρτον δὲ ἐπὶ τὸ ὀξύ ἐστι τὸ ἡμιτόνιον· δεύτερον δὲ οὗ τρίτον μὲν ἐπὶ τὸ βαρύ, πρῶτον δὲ ἐπὶ τὸ ὀξύ· τρίτον δὲ οὗ δεύτερον ἐφ' ἑκάτερα· τέταρτον δὲ οὗ πρῶτον μὲν ἐπὶ τὸ βαρύ, τρίτον δὲ ἐπὶ τὸ ὀξύ· πέμπτον δὲ οὗ τέταρτον μὲν ἐπὶ τὸ βαρύ, πρῶτον δὲ ἐπὶ τὸ ὀξύ· ἕκτον δὲ οὗ τρίτον μὲν ἐπὶ τὸ βαρύ, δεύτερον δὲ ἐπὶ τὸ ὀξύ· ἕβδομον δὲ οὗ δεύτερον μὲν ἐπὶ τὸ βαρύ, τρίτον δὲ

---

8 μέσων om. N. ἐπὶ M^4B (N in lit.), ἢ M^1W. 13 λοκριστί W, λοκροστί M^1, λοκρικὸν **B**M^3L. καὶ om. W. 14 δὲ τῷ δ. M^3. εἶδός ἐστι W. τὸ δ. MW**B**L, τοῦ δ. N. 15 οὗ] οὐ M. μὲν ἐπὶ τὸ βαρύ BM^3L, τε καὶ M^1W. δὲ ἐπὶ τὸ ὀξύ ἐστι] ἐπὶ M^1W. 16 τρίτον μὲν ἐπὶ τὸ βαρὺ om. M^1W. δὲ om. eid. 19 τρίτον] τέταρτον VWL, βαρύ, τρίτον δὲ ἐπὶ τὸ om. **B**. τέταρτον μὲν] πρῶτον M^1W. 20 πρῶτον δὲ] τέταρτον M^1, δεύτερον WL. πρῶτον δὲ ἑ. τὸ ὀξύ et τέταρτον μὲν ἑ. τὸ β. invertit B. 21 οὗ τρίτον μὲν ἑ. τὸ βαρὺ, δεύτερον δὲ M^4 in lit. et L, δεύτερον δὲ πέμπτον V, οὗ πρότερον μὲν ἑ. τὸ βαρὺ, δεύτερον δὲ W. 22 δεύτερον] πρότερον W. βαρὺ] ὀξὺ M^1W**B**. τρίτον] δεύτερον W, πέμπτον LM^4.

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ἐπὶ τὸ ὀξύ. ἔστι δὲ καὶ ταῦτα ἀπὸ τῶν αὐτῶν φθόγγων ἐπὶ τοὺς αὐτοὺς καθάπερ ἐπὶ τῆς ἁρμονίας καὶ τοῦ χρώματος καὶ ἐκαλεῖτο τοῖς αὐτοῖς ὀνόμασι.

10. Τῇ δὲ τοῦ ῥητοῦ καὶ ἀλόγου διαφορᾷ διοίσει συστήματα, ὅσα ἐκ ῥητῶν διαστημάτων σύγκειται, τῶν ἐξ ἀλόγων. ὅσα μὲν γὰρ ἐκ ῥητῶν ῥητά ἐστιν, ὅσα δὲ ἐξ ἀλόγων ἄλογα.

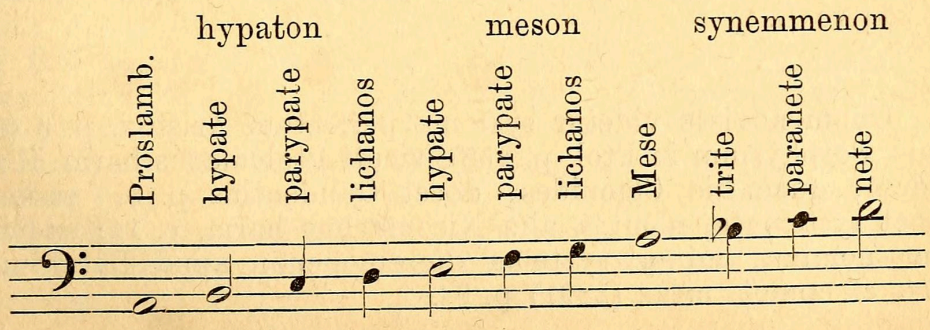
Τῇ δὲ τοῦ ἑξῆς καὶ ὑπερβατοῦ διαφορᾷ διοίσει (p. 17) συστήματα τὰ διὰ τῶν ἑξῆς φθόγγων μελῳδούμενα τῶν καθ' ὑπερβατόν.

Τῇ δὲ τοῦ συνημμένου καὶ διεζευγμένου διαφορᾷ διοίσει συστήματα, ὅσα διὰ τῶν συνημμένων τετραχόρδων τὴν σύνθεσιν ἔχει τῶν διὰ τῶν διεζευγμένων. ἔστι δὲ συναφὴ μὲν δύο τετραχόρδων ἑξῆς μελῳδουμένων, ὁμοίων κατὰ σχῆμα, φθόγγος κοινός. διάζευξις δέ ἐστι δύο τετραχόρδων ἑξῆς μελῳδουμένων, ὁμοίων κατὰ σχῆμα, τόνος ἀνὰ μέσον. εἰσὶ δὲ αἱ πᾶσαι συναφαὶ τρεῖς· μέση, ὀξυτάτη, βαρυτάτη. καὶ

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1 ὀξὺ] βαρύ W. καὶ om. WL, item l. 3 W. 5 τῶν] ἢ W. 6 γὰρ om. W. γὰρ ῥητὰ ἐκ ῥητῶν ἑ. libri. 9 τὰ inserit M¹, hab. NL, om. WB. 10 τῶν Bry. p. 385, τῇ libri. 12 διοίσει ὁμοίως σ. W. ὅσα] ἃ W. διὰ τῶν διέσεων συν. M¹WN. 13 τετρ.] τῶν τετρ. W. διὰ τῶν om. MNBL. 17 ἀνὰ μέσων N.

Systema synemmenon.



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